

where $|\mathcal{X}|$ denotes the cardinality of the input space. Second line follows from Corollary 1, which states that we have $|\mathcal{X}| = \mathcal{O}(2^{\alpha \cdot H(X)})$ for $\alpha \in [1, +\infty)$. Let f be a neural network with L intermediate representations forming a Markov chain:

$$X \rightarrow R_1 \rightarrow \dots \rightarrow R_L \rightarrow \hat{Y}$$

Fix any intermediate representation R_l , and decompose the network into $f = d \circ e$, where $e : \mathcal{X} \rightarrow \mathcal{R}_l$ maps inputs to $R_l = e(X)$, and $d : \mathcal{R}_l \rightarrow \mathcal{Y}$ predicts the output. Applying the generalization bound to the representation R_l , we obtain:

$$\mathcal{L}_{P(X,Y)}(f, \ell) = \mathcal{L}_{P(R_l,Y)}(d, \ell_e) \quad (3)$$

$$\leq \hat{\mathcal{L}}_{P(R_l,Y)}(d, \ell_e, \mathcal{D}_N) + \mathcal{O}\left(\frac{2^{\alpha \cdot H(R_l)} \log N}{\sqrt{N}}\right) \quad (4)$$

$$\leq \hat{\mathcal{L}}_{P(X,Y)}(f, \ell, \mathcal{D}_N) + \mathcal{O}\left(\frac{2^{\alpha \cdot H(R_l)} \log N}{\sqrt{N}}\right) \quad (5)$$

Since the Markov structure ensures that each R_l is a valid bottleneck in the information flow, we can take the tightest such bound across all layers, yielding:

$$\mathcal{L}_{P(X,Y)}(f, \ell) \leq \hat{\mathcal{L}}_{P(X,Y)}(f, \ell, \mathcal{D}_N) + \mathcal{O}\left(\frac{\log N \cdot \min_{1 \leq l \leq L} 2^{\alpha \cdot H(R_l)}}{\sqrt{N}}\right)$$

This concludes the proof. $\square$

Theorem 2. *The Information Bottleneck Language Modeling (IBLM) objective defined in Equation 2 is equivalent to the classical Information Bottleneck formulation under the language modeling setting.*

Proof. The Information Bottleneck (IB) framework [2, 8] defines the optimal representation R as the solution to:

$$\min_{p(r|x)} I(R; X) \quad (6)$$

$$\text{s.t. } I(Y; R) = I(Y; X) \quad (7)$$

That is, R should discard as much input information as possible while preserving all information relevant to predicting Y —the minimal sufficient statistic. In large language models (LLMs), the network forms a deterministic Markov chain: $X \rightarrow R \rightarrow \hat{Y}$. Since R is deterministically computed from X , we have:

$$I(X; R) = H(R) \quad (\text{as } H(R|X) = 0) \quad (8)$$

$$I(Y; X) \geq I(Y; R) \geq I(Y; \hat{Y}) = H(Y) - H(Y|\hat{Y}) \quad (9)$$

Where second line follows from data processing inequality [3]. Thus,

$$I(Y; X) - I(Y; R) \leq H(Y|\hat{Y}) - H(Y|X) \quad (10)$$

Since $H(Y|X)$ is fixed, minimizing $H(Y|\hat{Y})$ —i.e., cross-entropy—maximizes $I(Y; \hat{Y})$, satisfying the IB predictive constraint. On the compression side, since $I(X; R) = H(R)$, minimizing representation entropy directly reduces the IB compression term. Hence, minimizing cross-entropy aligns with preserving predictive information, while minimizing entropy enforces compression—together forming the IBLM objective. $\square$

3 Approach

While applying a Lagrangian objective (CE + λ ·MBE) is a natural approach to solving the constrained optimization in IBLM, we find it often leads to representation collapse: MBE converges to near-zero, but CE worsens as the model loses structure in its internal representations.

Inspired by the alternation between learning and consolidation in biological systems, we divide training into two phases: memorization, where the model minimizes cross-entropy (CE) loss, and compression, where it minimizes a weighted sum of CE and Matrix-Based Entropy (MBE). We

propose Gated Phase Transition (GAPT), a training algorithm that dynamically alternates between these phases. GAPT tracks a global minimum CE loss and per-layer MBE histories, and uses patience-based gating to switch phases. Compression is exited early if CE degrades.

GAPT encourages localized compression—reorganizing existing knowledge without interfering with the acquisition of new information—and ensures that entropy reduction occurs only when it does not hinder memorization.

Algorithm 1 Gated Phase Transition (GAPT)

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1: Input: losses  $\mathcal{L}$ , thresholds  $\delta, \tau$ , patience  $p_m, p_c$ 
2: State:  $\phi \in \{1, 2\}$  (1 = mem, 2 = comp), counters  $s_m, s_c, E_{\min}, \text{MBE}_{\min}[i]$ 
3: Extract  $\mathcal{L}_{\text{ce}}, \{\text{MBE}_i\}$ ; update  $\Delta E \leftarrow E_{\min} - \mathcal{L}_{\text{ce}}, E_{\min} \leftarrow \min(E_{\min}, \mathcal{L}_{\text{ce}})$ 
4: if  $\phi = 1$  then ▷ Memorization
5:    $s_m \leftarrow 0$  if  $\Delta E > \delta$  else  $s_m += 1$ 
6:   if  $s_m \geq p_m$  then
7:      $\phi \leftarrow 2, s_c \leftarrow 0, E_{\min} \leftarrow \infty, \text{MBE}_{\min}[i] \leftarrow \infty$ 
8:   end if
9: else ▷ Compression
10:  if  $\mathcal{L}_{\text{ce}} > E_{\min} \cdot (1 + \tau)$  then
11:     $\phi \leftarrow 1, s_m \leftarrow 0$ 
12:  else
13:     $\Delta M \leftarrow \max_i(\text{MBE}_{\min}[i] - \text{MBE}_i)$ 
14:    for each  $i$  do
15:       $\text{MBE}_{\min}[i] \leftarrow \min(\text{MBE}_{\min}[i], \text{MBE}_i)$ 
16:    end for
17:     $s_c \leftarrow 0$  if  $\Delta M > \delta$  else  $s_c += 1$ 
18:    if  $s_c \geq p_c$  then
19:       $\phi \leftarrow 1, s_m \leftarrow 0$ 
20:    end if
21:  end if
22: end if

```

4 Experiments

4.1 Natural Compression-Memorization cycle

Our experimental setup follows the Modded-NanoGPT framework [23]. We remove FP8 matmul (due to hardware incompatibility with Hopper GPUs) and use a simplified block causal attention mask. All experiments are conducted on 8xL40 GPUs, training a 12-layer GPT model on a 0.73B-token FineWeb training set, evaluated on its corresponding validation split. We train on CE loss only. We log cross-entropy (CE) and Matrix-Based Entropy (MBE) gradients at every training step. For each iteration, we record both CE and MBE gradients across multiple batches and compute the average cosine similarity both (1) across batches (CE vs. CE), and (2) between CE and MBE gradients, within and across batches.

In Figure 1, we observe that training with CE loss alone leads to a consistent decrease in MBE across layers, confirming observations from [14]. We further analyze CE gradient behavior. Figure 2. shows that gradient consistency (measured by cosine similarity across batches) declines over time across all layers. This indicates an increasing signal-to-noise ratio in CE gradients as training progresses.

We next inspect cosine similarity between CE and MBE gradients. As shown in Figure 3, we observe recurring sign flips that indicate an alternation between memorization and compression phases, even without explicitly optimizing for entropy. To characterize this oscillation, we analyze the gradient signal across training using three measures: standard deviation (oscillation strength), zero-crossing rate (oscillation frequency), and peak-to-average power ratio from the power spectral density (periodicity strength). Figure 4 summarizes these metrics across layers and parameters. We find that attention parameters exhibit stronger and more frequent oscillations than MLP parameters. Interestingly, later layers show higher oscillation frequency, while no layer demonstrates strong

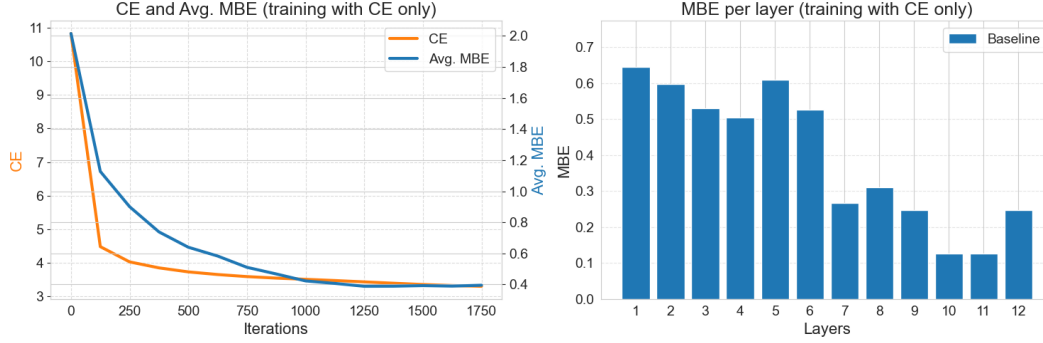


Figure 1: Left: CE and MBE loss curves during pretraining with CE loss only, showing implicit momentum for representation compression. Right: final per-layer MBE values. Later layers show lower MBE, indicating representation compression.

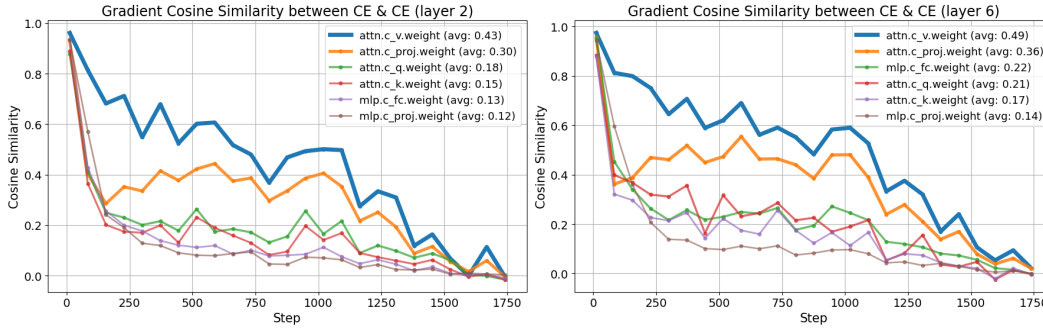


Figure 2: Cosine similarity between CE gradients across batches. CE gradients become increasingly decorrelated over time, reflecting diminishing shared signal.

rhythmic periodicity—suggesting that oscillation is irregular and state-driven, rather than strictly periodic.

4.2 Pre-training with GAPT

In our second experiment, we retain the same GPT pre-training setup but incorporate the GAPT algorithm to test whether it offers a better solution to IBLM objective defined in Equation 2, compared to a baseline model trained solely on the CE loss. MBE regularization during compression phase in GAPT is done from layer 2 to 9.

Table 1: Cross-entropy loss on the FineWeb validation set.

Model	CE Loss
Baseline	3.31
GAPT (Ours)	3.15 (-4.8%)

As shown in Table 1, GAPT reduces cross-entropy on the FineWeb validation set by 4.8% compared to the CE-only baseline. In addition to reducing test CE loss, GAPT also significantly compresses internal representations. Figure 5 (left) shows the layer-wise MBE for both models. We observe consistent reductions across all regularized layers. The per-layer MBE values and their relative improvements are summarized in Figure 5 (right). GAPT reduces MBE by an average of 70.5% across layers 2–9 while improving validation cross-entropy by 4.8%. This suggests that explicitly alternating between memorization and compression phases offers an effective solution to the